

# **FOREVER – FULL STACK E-COMMERCE WEBSITE**

## **PROJECT REPORT**

### **ABSTRACT**

FOREVER is a modern full-stack e-commerce web application developed using the MERN Stack (MongoDB, Express.js, React.js, and Node.js). The platform enables users to browse products, register accounts, manage shopping carts, place orders, and track purchases. An integrated admin panel allows administrators to manage products, users, and orders efficiently.

The system incorporates secure authentication using JWT, password encryption through bcrypt, image storage via Cloudinary, and a scalable MongoDB database architecture. The project demonstrates the practical implementation of full-stack development concepts and modern web technologies.

Keywords: MERN Stack, E-Commerce, MongoDB, React, Node.js, Express.js, JWT, Cloudinary.

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## **CHAPTER 1: INTRODUCTION**

### **1.1 Background**

E-commerce has transformed traditional shopping by enabling users to purchase products online. Businesses increasingly rely on web-based platforms to reach customers worldwide.

### **1.2 Need for the System**

Traditional shopping systems suffer from:

- Geographical limitations
- Limited operating hours
- Manual inventory management
- Poor scalability

FOREVER addresses these issues through a digital shopping platform.

### **1.3 Objectives**

- Develop a responsive e-commerce platform
- Provide secure authentication
- Manage products efficiently

- Enable online order placement
  - Provide administrative control
  - Ensure scalability and security
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## **CHAPTER 2: LITERATURE SURVEY**

### **Existing Systems**

#### **Amazon**

Large-scale online shopping platform.

#### **Flipkart**

Indian e-commerce marketplace.

#### **Myntra**

Fashion-focused online store.

### **Limitations**

- High development complexity
- Expensive customization
- Complex deployment

### **Proposed System**

FOREVER provides:

- Lightweight architecture
  - MERN Stack implementation
  - Cloud-based image management
  - Secure user authentication
  - Responsive user interface
- 

## **CHAPTER 3: SYSTEM ANALYSIS**

### **Functional Requirements**

### **User Module**

- Registration
- Login
- Product browsing
- Cart management
- Order placement

### **Admin Module**

- Product management
- Order management
- Inventory monitoring

### **Non-Functional Requirements**

- Security
- Scalability
- Reliability
- Maintainability
- Performance

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## **CHAPTER 4: SYSTEM DESIGN**

### **Architecture**

User

↓

Frontend (React)

↓

Backend API (Node + Express)

↓

MongoDB Database

↓

Cloudinary Storage

### **Data Flow**

1. User requests product data
  2. Frontend calls APIs
  3. Backend processes request
  4. MongoDB returns data
  5. Frontend displays results
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## **CHAPTER 5: TECHNOLOGY STACK**

### **Frontend**

#### **React.js**

Component-based UI development.

#### **React Router**

Client-side routing.

#### **Axios**

HTTP request handling.

#### **Tailwind CSS**

Responsive styling.

#### **React Toastify**

Notification management.

#### **Vite**

Fast development environment.

### **Backend**

#### **Node.js**

Server-side runtime environment.

#### **Express.js**

REST API framework.

#### **MongoDB**

NoSQL database.

### **Mongoose**

MongoDB object modeling.

### **Security**

#### **JWT**

Authentication tokens.

#### **bcrypt**

Password hashing.

### **Media Storage**

#### **Cloudinary**

Image upload and storage.

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## **CHAPTER 6: PROJECT STRUCTURE**

### **Admin Folder**

- src/
- public/
- package.json
- vite.config.js

Functions:

- Product Management
- Order Monitoring
- Admin Dashboard

### **Backend Folder**

#### **Config**

mongodb.js  
cloudinary.js

## **Controllers**

userController.js  
productController.js  
cartController.js  
orderController.js

## **Middleware**

auth.js  
adminAuth.js  
multer.js

## **Models**

userModel.js  
productModel.js  
orderModel.js

## **Routes**

API route management

## **Server**

server.js

## **Frontend Folder**

### **Components**

Reusable UI components.

### **Pages**

- Home
- Collection
- Product
- Cart
- Login
- Orders

## **User Collection**

Fields:

- \_id
- name
- email
- password
- cartData

## **Product Collection**

Fields:

- \_id
- name
- description
- category
- subCategory
- price
- image
- sizes

## **Order Collection**

Fields:

- \_id
- userId
- items
- amount
- address
- paymentMethod
- status

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## **CHAPTER 8: MODULE DESCRIPTION**

### **Authentication Module**

Features:

- Registration
- Login
- JWT generation
- Password encryption

### **Product Module**

Features:

- Add products
- Update products
- Delete products
- View products

### **Cart Module**

Features:

- Add to cart
- Remove from cart
- Quantity update

### **Order Module**

Features:

- Place orders
- Order history
- Status updates

### **Admin Module**

Features:



- Product management
  - Order tracking
  - Administrative control
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## **CHAPTER 9: API DOCUMENTATION**

### **User APIs**

POST /register

POST /login

### **Product APIs**

GET /list

POST /add

POST /remove

### **Cart APIs**

POST /add

POST /update

GET /get

### **Order APIs**

POST /place

POST /verify

GET /userorders

GET /list

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## **CHAPTER 10: SECURITY IMPLEMENTATION**

### **JWT Authentication**

Secure token generation and verification.

### **Password Hashing**

bcrypt hashing protects user credentials.

### **Authorization**

Role-based admin access.

### **Environment Variables**

Sensitive credentials stored securely.

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## **CHAPTER 11: TESTING**

### **Unit Testing**

Individual modules tested separately.

### **Integration Testing**

Communication between modules verified.

### **System Testing**

Complete workflow tested.

### **User Acceptance Testing**

Real-world functionality validated.

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## **CHAPTER 12: RESULTS**

Successfully implemented:

- ✓ User Authentication
- ✓ Product Management
- ✓ Shopping Cart
- ✓ Order Processing
- ✓ Admin Dashboard
- ✓ Cloudinary Integration
- ✓ MongoDB Database Connectivity

## **CHAPTER 13: FUTURE SCOPE**

- AI Product Recommendations
  - Product Reviews
  - Wishlist
  - Coupons
  - Online Payment Gateway Expansion
  - Multi-Vendor Marketplace
  - Mobile Application
  - Advanced Analytics Dashboard
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## **CHAPTER 14: CONCLUSION**

FOREVER successfully demonstrates the development of a modern full-stack e-commerce platform using the MERN Stack. The system provides secure authentication, efficient product management, seamless shopping experience, and scalable architecture suitable for real-world deployment.

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## **REFERENCES**

1. React.js Documentation
2. Node.js Documentation
3. Express.js Documentation
4. MongoDB Documentation
5. Mongoose Documentation
6. JWT Documentation
7. Cloudinary Documentation
8. Tailwind CSS Documentation

## **CHAPTER 15: IMPLEMENTATION DETAILS**

### **15.1 Home Page**

#### **Description**

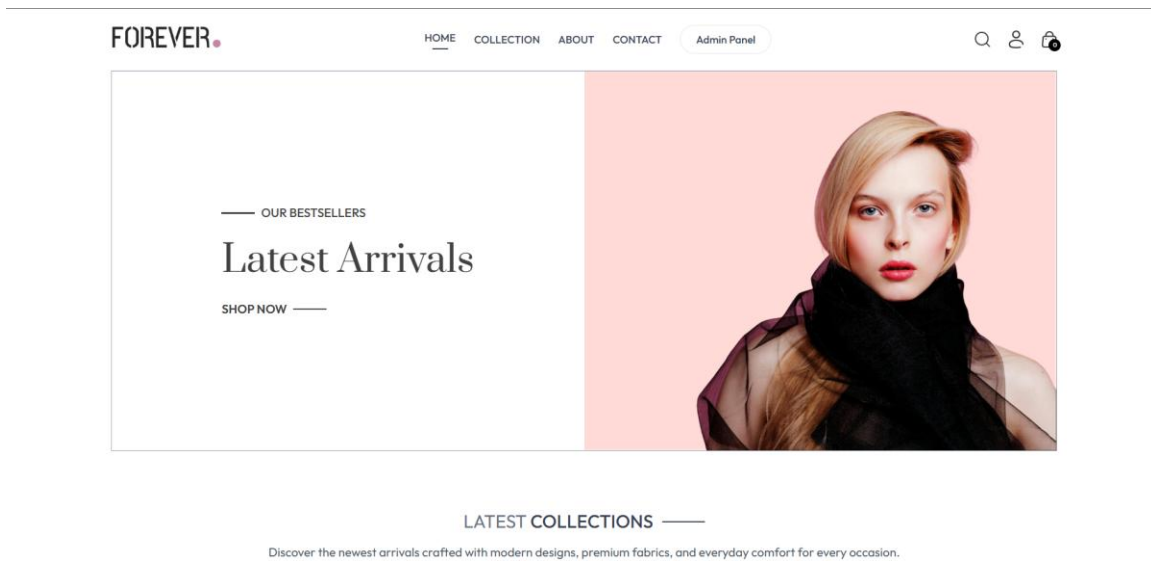
The Home Page serves as the landing page of the FOREVER E-Commerce Platform.

#### **Features Implemented**

- Responsive Navigation Bar
- Company Logo (FOREVER)
- Product Search Icon
- User Authentication Icon
- Shopping Cart Icon
- Admin Panel Access
- Hero Banner Section
- Latest Collections Section
- Best Sellers Section
- Newsletter Subscription
- Footer Section

#### **Screenshot**

**Figure 15.1: Home Page**



## Working

The homepage displays featured products and latest arrivals. Users can navigate through product categories and access various functionalities such as search, login, cart management, and admin panel.

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## 15.2 Collection Page

### Description

The Collection Page displays all available products in the store.

### Features Implemented

#### Category Filters

- Men
- Women
- Kids

#### Product Type Filters

- Topwear
- Bottomwear
- Winterwear

#### Product Sorting

- Relevant
- Low to High Price
- High to Low Price

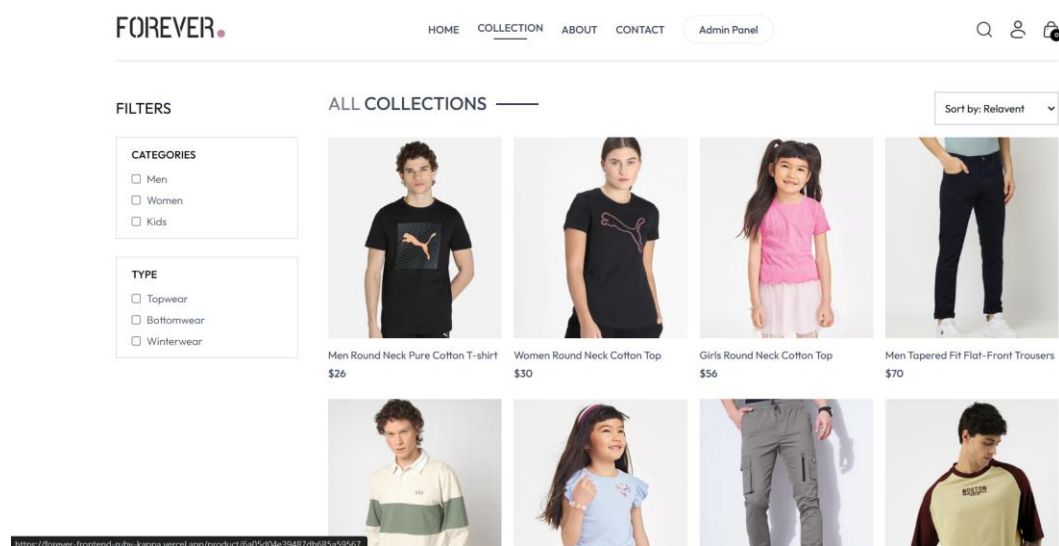
## Product Cards

Each card contains:

- Product Image
- Product Name
- Product Price
- Product Details

## Screenshot

**Figure 15.2: Collection Page**



## Working

Users can filter products based on categories and clothing types. Products are dynamically fetched from MongoDB through backend APIs.

## 15.3 About Us Page

### Description

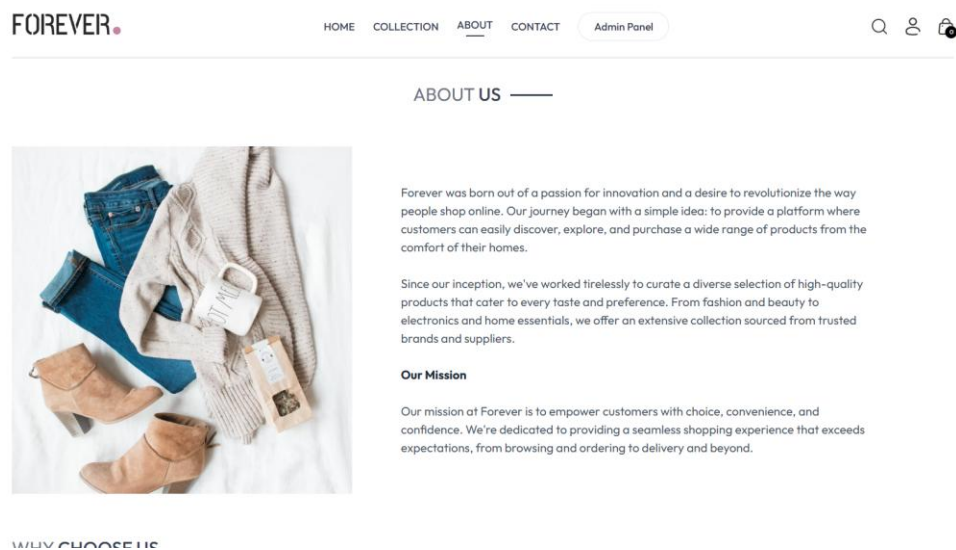
The About Us page provides company information and mission statement.

## Features

- Company Introduction
- Business Vision
- Mission Statement
- Brand Story
- Customer Commitment

## Screenshot

**Figure 15.3: About Us Page**



## Working

This page explains the purpose of the FOREVER platform and its commitment to providing quality products and excellent customer service.

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## 15.4 Contact Us Page

### Description

The Contact Page provides communication details.

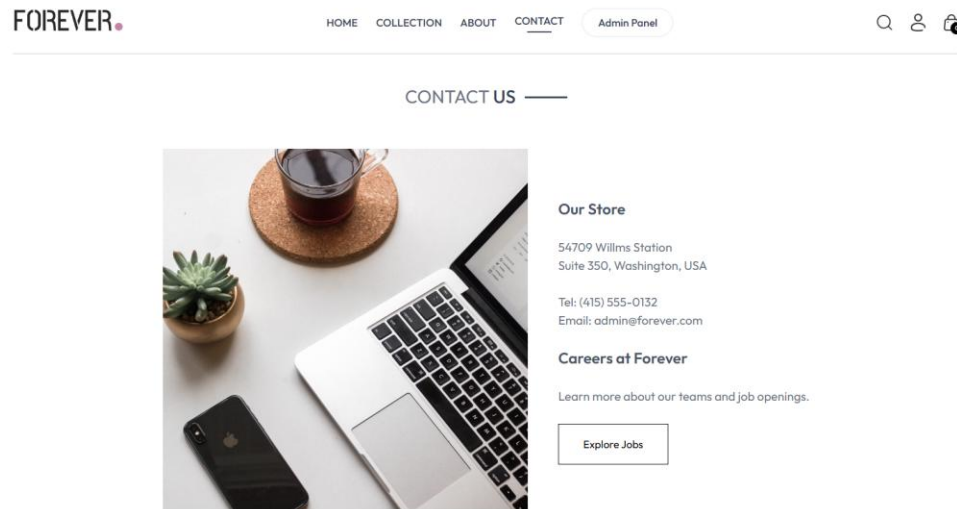
### Features

- Store Address
- Phone Number

- Email Address
- Career Information
- Explore Jobs Button

## Screenshot

**Figure 15.4: Contact Us Page**



## Working

Users can obtain company contact details and career-related information through this page.

## CHAPTER 16: USER INTERFACE DESIGN

### Navigation Bar

The navigation bar contains:

#### Component Purpose

|            |                       |
|------------|-----------------------|
| Home       | Redirects to Homepage |
| Collection | Displays Products     |
| About      | Company Information   |
| Contact    | Contact Information   |



## Component Purpose

Admin Panel Administrator Dashboard

Search      Product Search

User      Login/Profile

Cart      Shopping Cart

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## Hero Section

The Hero Section includes:

- Promotional Banner
- Latest Arrivals Advertisement
- Call-To-Action Button
- Fashion Product Showcase

Purpose:

- Increase customer engagement
  - Highlight featured products
  - Improve conversion rate
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## Product Listing Design

Each product card contains:

- Product Image
- Product Name
- Product Price
- Clickable Product Details

Benefits:

- Clean User Experience

- Easy Product Navigation
- Responsive Layout

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## CHAPTER 17: DATABASE IMPLEMENTATION

### MongoDB Collections

#### Users Collection

```
{
  "_id": "",
  "name": "",
  "email": "",
  "password": "",
  "cartData": {}
}
```

#### Products Collection

```
{
  "_id": "",
  "name": "",
  "description": "",
  "price": "",
  "category": "",
  "subCategory": "",
  "sizes": [],
  "image": []
}
```

#### Orders Collection

```
{
  "_id": "",
  "userId": "",
  "items": [],
  "amount": "",
  "address": {},
  "status": ""
}
```

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## CHAPTER 18: DEPLOYMENT DETAILS

## **Frontend Deployment**

Platform:

Vercel

Frontend URL:

<https://forever-frontend-ruby-kappa.vercel.app>

### **Advantages**

- Fast Deployment
  - Global CDN
  - Automatic SSL
  - GitHub Integration
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## **Backend Deployment**

Technology:

- Node.js
- Express.js

Deployment Platform:

- Vercel / Render (as configured)
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## **Database Hosting**

Platform:

MongoDB Atlas

Benefits:

- Cloud Database
- Scalability
- Backup Support
- Secure Access

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## CHAPTER 19: PROJECT ACHIEVEMENTS

Successfully Implemented:

- ✓ MERN Stack Architecture
- ✓ Responsive UI Design
- ✓ Product Management
- ✓ Authentication System
- ✓ Shopping Cart
- ✓ Order Management
- ✓ MongoDB Integration
- ✓ Cloudinary Image Upload
- ✓ Admin Dashboard
- ✓ REST API Development
- ✓ Frontend Deployment
- ✓ Backend Deployment

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## CHAPTER 20: CONCLUSION

The FOREVER Full Stack E-Commerce Website is a complete MERN-stack application developed to provide a modern online shopping experience. The project successfully integrates React.js for frontend development, Node.js and Express.js for backend services, MongoDB for database management, and Cloudinary for image storage. The application provides a responsive user interface, secure authentication system, product management, order processing, and administrative functionalities.

The successful deployment of the project demonstrates practical implementation of full-stack development concepts and industry-standard software engineering practices.

### Technologies Used

- React.js
- Vite

- Tailwind CSS
- Node.js
- Express.js
- MongoDB Atlas
- Mongoose
- JWT Authentication
- bcrypt
- Cloudinary
- Axios
- Vercel

### **Final Outcome**

A fully functional online fashion e-commerce platform capable of handling real-world product browsing, shopping cart management, user authentication, and order processing operations.